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Research paper

Fault classification and localization in power transmission line based on machine learning and combined CNN-LSTM models

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ABSTRACT

Accurate classification and localization of faults in power transmission lines are critical for maintaining grid stability and minimizing downtime. However, it remains challenging due to the variability of fault conditions and system complexities. In this paper, we present a novel method to classify and localize faults in power transmission line using machine learning algorithms. The IEEE 9 Bus is simulated in MATLAB Simulink and faults are generated with various conditions including fault types, load level, fault resistance, and fault location. More than 300 thousand fault datasets are created for both classifying and localizing tasks. Support Vector Machine (SVM), Decision Tree (DT), Logistic Regression (LR), XGBoost, and Artificial Neural Network (ANN) are used for fault classification while Convolutional Neural Network (CNN), Long-Short Term Memory (LSTM) and combined CNN-LSTM models are used for fault localization. The results show that XGBoost can classify fault with the accuracy of 99.82 %, while the CNN-LSTM can localize fault with MAPE $<$ 1 % and MAE $<$ 0,16 km.

1. Introduction

The rapid development of the economy and society has led to the need for increased power consumption and reliability. Building and managing a reliable power system has become essential to meet those demands. Due to the increasing power consumption, the power transmission line tends to work at their maximum capacity. Faults at power transmission lines often lead to system outages and affect millions of residents. Among massive record blackout events in the world, the 30th July 2012 blackout in India affected over 400 million people and was briefly the largest power outage in history (Chakraborty et al., 2015; Wu et al., 2017). The 400 kV line circuit breaker tripped, causing all major stations in affected states to shut down. The main reason was due to the increased power transfer which exceeded the transmission line capacity. Another major blackout occurred on 24th August 2003 that affected as many as 50 million people in many big cities in the USA and Canada. The short circuit due to a tree getting into contact with one of the overhead lines was determined as the cause. This massive blackout's damage was estimated to be 6 billion USD. Faults in transmission lines normally cause severe blackouts and affect large residential areas. Therefore, early fault detection in transmission line is critical to prevent further system outages. There are numerous studies about fault detection and localization in power transmission lines. In convention, protective relays at both ends of the transmission line can detect and isolate the faults immediately. Distance relays are widely used to detect and localize the line-line and line-ground fault position in transmission line (Idris et al., 2013; Mahari and Sanaye-Pasand, 2017; Lin et al., 2023). However, distance relay's accuracy may be affected by several factors such as fault resistance, overreach/underreach condition, power swing, CT saturation, Wavelet transform is a powerful tool for detecting and localizing faults in transmission lines due to its ability to analyze both the time and frequency components of signals (Banerjee et al., 2022; Shaik and Pulipaka, 2015; Ahmed et al., 2023; Ekici et al., 2008; Aparicio et al., 2021; Chen et al., 2022). Wavelet transform decomposes a signal into wavelet coefficients, which provide a time-frequency representation of the signal. Unlike the Fourier transform, which only provides frequency information, wavelet transform retains time information, making it particularly useful for analyzing transient signals such as those caused

Abbreviations: ANN,, Artificial Neural Network; CNN,, Convolutional Neural Network; DT,, Decision Tree; GSM,, Global System for Mobile Communications; LR,, Logistic Regression; LSTM,, Long-Short Term Memory; MAE,, Mean Absolute Error; MAPE,, Mean Absolute Percentage Error; MRA,, Multi-Resolution Analysis; MSE,, Mean Squared Error; RELU,, Rectifier Linear Unit; RNN,, Recurrent Neural Network; SGD,, Stochastic Gradient Descent; SVM,, Support Vector Machine; WSN,, Wireless Sensor Network; XGBOOST,, eXtreme Gradient Boosting.

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by faults in transmission lines. To detect faults, the wavelet transform is applied to the voltage and current signals from the transmission line. Once a fault is detected, its location can be determined using Energy of Wavelet Coefficients (Ekici et al., 2008), Multi-Resolution Analysis (MRA) (Aparicio et al., 2021), or Travelling Wave Method (Chen et al., 2022). Fuzzy logic is another effective technique for detecting and localizing faults in power systems, especially where the system behavior is complex and uncertain (Bhatnagar and Yadav, 2020; Peng et al., 2018; Adhikari et al., 2016). Fuzzy logic is a form of multi-valued logic with an approximation approach. Unlike classical binary sets (where variables are either 0 or 1), fuzzy logic variables may have a truth value that ranges between 0 and 1. This makes fuzzy logic suitable for handling uncertainty and imprecision, which are common in fault detection. The combined wavelet transforms, and fuzzy logic is also used to enhance fault detection and localization in transmission lines (Jamil et al., 2015; Bunjongjit et al., 2013). Communication networks such as Global System for Mobile Communications (GSM) and Wireless Sensor Network (WSN) are also widely used to detect and localize faults in transmission lines (Gulbhile et al., 2017; Gunasekar et al., 2021; Sujatha and Kumar, 2011; Raja et al., 2023; Leoni et al., 2014; Li et al., 2016). In GSM technology, the remote devices can be programmed to detect abnormal conditions and immediately send an alert to the central monitoring system via GSM. They can include GPS modules to provide precise location data. When a fault is detected, the device sends its GPS coordinates along with the fault data (Gulbhile et al., 2017; Gunasekar et al., 2021; Sujatha and Kumar, 2011). In WSN technology, sensors are strategically placed along the transmission line to measure parameters like voltage, current, temperature, and vibration. Sensors can perform local processing to detect sudden changes or patterns that indicate a fault. For example, a significant drop in voltage or spike in current can trigger an alert. In addition, by correlating data from different sensors, the system can estimate the fault location (Raja et al., 2023; Leoni et al., 2014; Li et al., 2016). For instance, the time difference in the detection of a fault by different sensors can be used to estimate the fault's position. Recently, artificial intelligence, machine learning and deep learning have been widely used in power system control, management, and optimization, leading to enhanced reliability, efficiency, and sustainability. Applications include demand forecasting (Huy et al., 2022; Ahmad et al., 2022), energy management and optimization (Luo et al., 2023), microgrids (Mohammadi et al., 2022), predictive maintenance (Ucar et al., 2024). These techniques enable machines to simulate human intelligence, including decision-making, problem-solving, and allow systems to learn and make predictions based on data. Traditional fault detection and localization methods, such as impedance-based techniques and traveling wave methods mentioned above, often struggle with complex network topologies, high impedance faults, and varying operating conditions. Additionally, manual analysis and diagnosis can be time-consuming and prone to human error. With the advancement of data acquisition technologies and the increasing availability of large datasets from power systems, there is a growing interest in utilizing machine learning algorithms for fault analysis. ML techniques can automatically learn from data, identify patterns, and make predictions, offering a promising approach to enhance fault detection and localization in power transmission lines. ML techniques usually achieve higher accuracy, faster response times, and better adaptability to varying system conditions. The main contribution of the paper is described below:

- Development of an ML-based fault classification and localization framework: The paper presents a novel framework utilizing various machine learning algorithms for the classification of faults in power transmission lines. The proposed framework is capable of distinguishing between different types of faults, such as single line-to-ground faults, line-to-line faults, and double line-to-ground faults, among others. In addition to fault classification, the paper introduces recurrent neural network models for accurate fault localization. This allows for the precise identification of the fault's location along the transmission line, enabling faster repair and restoration efforts.
- Comprehensive analysis and comparison of ML algorithms: The paper provides an extensive analysis of different ML algorithms, including Support Vector Machine (SVM), Decision Tree (DT), Logistic Regression (LR), XGBoost, and Artificial Neural Network (ANN), Convolutional Neural Network (CNN), Long-Short Term Memory (LSTM) and combined CNN-LSTM. It compares their performance in terms of accuracy, computational efficiency, offering valuable insights into the most effective techniques for fault analysis in power systems.
- Dataset creation and utilization: The paper involves the creation and utilization of a comprehensive simulated fault scenarios dataset. This dataset is used to train and validate the ML models, ensuring that the findings are applicable to practical power transmission systems.

The paper is structured as follows: Section I introduces the literature review, and research goal; Section II focuses on classification and regression models; In section III, the machine learning algorithms are applied to fault datasets generated from IEEE 9 Bus system; Finally, conclusions and future research directions are provided in section IV.

2. Fault classification and localization models

Machine learning is mostly used for classification and regression problems. In this paper, machine learning models such as Support Vector Machine (SVM), Decision Tree (DT), Logistic Regression (LR), XGBoost, and Artificial Neural Network (ANN) are used for fault classification (classification problems), while Convolutional Neural Network (CNN), Long-Short Term Memory (LSTM) and combined CNN-LSTM models are used for fault localization (regression problems).

2.1. Artificial neural network

ANN model is one of the most common choices for classification problems. The network structure consists of 3 layers with the number of neurons being optimized for each layer (Fig. 1). The first layer is known as the input layer which represents input data. The output layer consists of neurons which predict the output of the model. Hidden layers adjust weights and biases to minimize the error between predicted and actual outputs. Each neuron produces a signal from the previous layer through multivariate linear regression and activation function. Three common types of activation function are sigmoid, hyperbolic tangent and rectifier linear unit (ReLU). The ReLU (Fig. 2) is more preferred since it overcomes the vanishing gradient problem of the sigmoid and hyperbolic function and allows the network to learn faster.

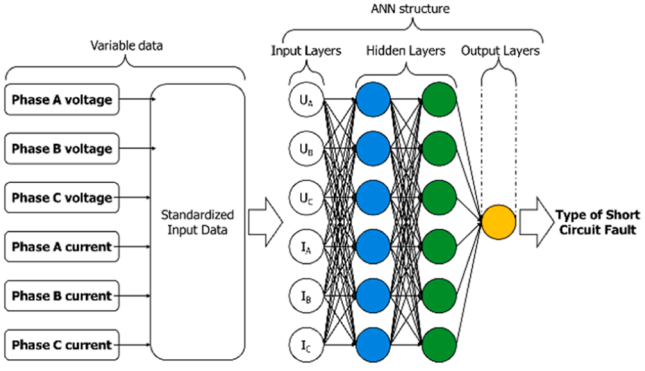


Fig. 1. Artificial neural network model.

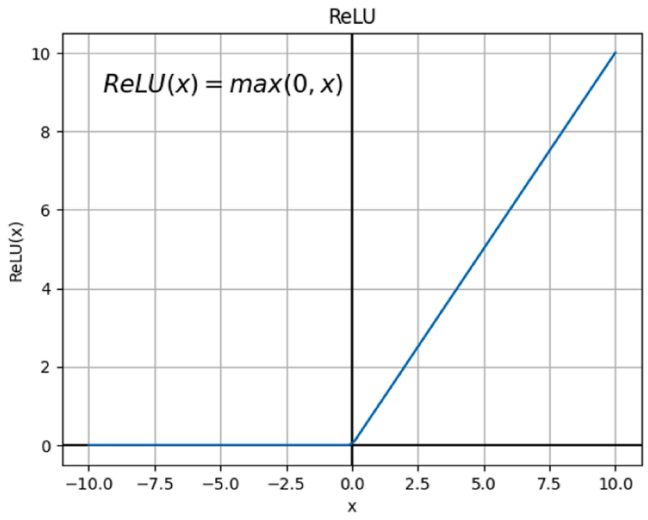


Fig. 2. Rectifier Linear Unit function.

2.2. Decision tree

Decision tree is a supervised learning algorithm used in both classification and regression problems. DT algorithm usually have high accuracy for classification problems with non-linear database. DT consists of several components: root node, decision node and leaf node (Fig. 3). DT starts from the root node, compares the root attribute with the record's attribute, then jumps to the next node based on the comparison.

For classification problems, DT uses Gini index, Entropy, and Information Gain (Aurélien, 2022):

- Gini Index:

$$G(t) = 1 - \sum_{i=1}^{K} p_i^2 \tag{1}$$

where p_i is the proportion of samples belonging to class i at node t.

- Entropy:

$$H(t) = -\sum_{i=1}^{K} p_i \log_2(p_i) \tag{2}$$

Information Gain:

$$IG(A) = H\left(parent\right) - \sum_{j=1}^{m} \frac{n_j}{n} H(t_j) \tag{3}$$

where:

H(parent) is the entropy of the parent node.

$H(t_j)$ is the entropy of child node j.

n_j is the number of samples in child node j.

n is the total number of samples in the parent node.

2.3. Support vector machine

SVM is one of the most robust classification algorithms. SVM is widely used because of its flexibility to classify both linear and nonlinear data structure. In SVM, the data are represented in an-dimension chart (Fig. 4). SVM can ignore abnormal data to find the optimal boundary for classification.

For linear data, a hyperplane is defined by a weight vector w and bias b (Aurélien, 2022):

$$w.x + b = 0 \tag{4}$$

The decision function for classifying a new data point x is:

$$f(x) = sign(w.x + b) \tag{5}$$

For non-linear data, SVM uses a kernel function $K(x_i,x)$ to map the data to a higher-dimensional space where a linear hyperplane can separate the classes:

$$f(x) = sign\left(\sum_{i=1}^{n} \alpha_i y_i K\left(x_i, x\right) + b\right) \tag{6}$$

2.4. Regression models

Linear regression and logistic regression are the most common type of regression and are widely used in supervised learning problems (Fig. 5). The main different between these two regression models is that linear regression is used to predict the continuous dependent variable from the independent variables, while the logistic regression is used to predict categorical dependent variables from the independent variables. Therefore, linear regression is more suitable for solving regression problems while logistic regression is more preferred to solve the classification problems.

In logistic regression, the linear combination $z = w.x + b$ is passed through a sigmoid function to map it to a probability value between 0 and 1:

$$\sigma(z) = \frac{1}{1 + e^{-z}} \tag{7}$$

The sigmoid function, also known as the logistic function, ensures that the output is a valid probability. The probability that the input x belongs to class 1 is given by:

$$P\left(y = 1|x\right) = \sigma\left(w.x + b\right) = \frac{1}{1 + e^{-(w.x+b)}} \tag{8}$$

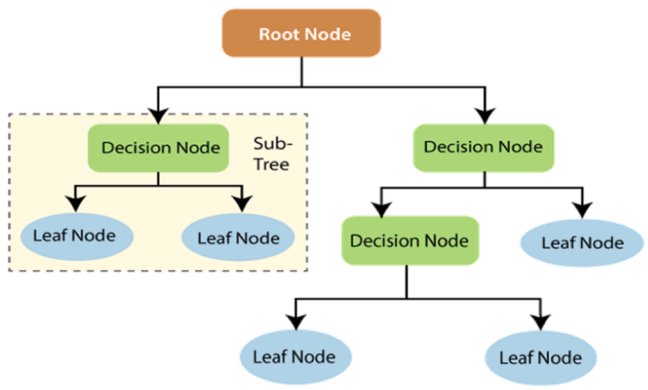


Fig. 3. Decision tree model.

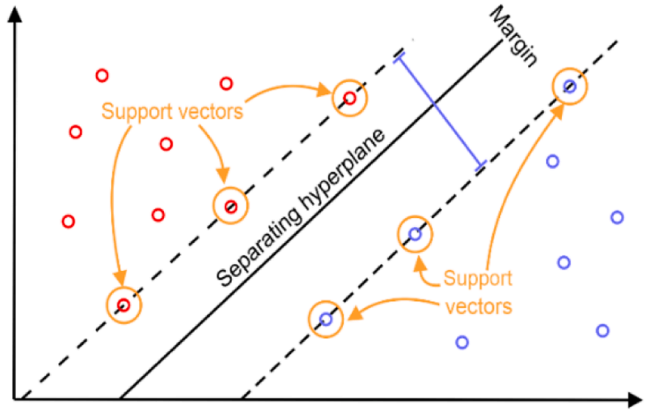


Fig. 4. Support vector machine model.

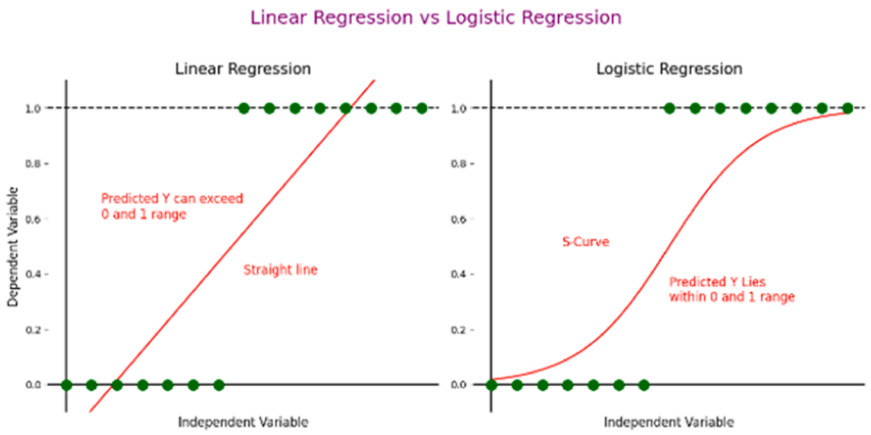


Fig. 5. Linear regression (left) and logistic regression (right).

The probability that the input x belongs to class 0 is:

$$P(y=0|x) = 1 - P(y=1|x) \tag{9}$$

2.5. XGBoost

XGBoost (eXtreme Gradient Boosting) is an advanced machine learning algorithm based on gradient boosting that is highly effective for classification problems (Fig. 6). XGBoost builds an ensemble of weak learners (typically decision trees) in a sequential manner. Each new tree is trained to correct the errors of the combined ensemble of previous trees. The model makes predictions by combining the outputs of multiple trees. XGBoost uses an objective function that includes both a loss function to measure model performance and a regularization term to prevent overfitting:

$$L = \sum_{i=1}^{n} l\left(y_i, \hat{y}_i\right) + \sum_{k=1}^{K} \Omega\left(f_k\right) \tag{10}$$

- Loss function (l): Measures the difference between the actual and predicted values. Common loss functions include mean squared error for regression and logistic loss for classification.

- Regularization Term (Ω): Penalizes the complexity of the model, which is typically measured by the number of leaves in the tree and the magnitude of the leaf weights:

$$\Omega(f_k) = \gamma T + \frac{1}{2}\lambda \sum_{j=1}^{T} w_j^2 \tag{11}$$

where T is the number of leaves in the k-th tree, w_j is the weight of leaf j, γ controls the number of leaves, and λ controls the L2 regularization.

2.6. Convolutional neural network

CNNs are typically associated with classification problems, particularly in image recognition, but they can also be effectively used for regression problems. The architecture of a CNN used for regression problems is like classification problems, but with key differences, particularly in the output layer (Cui and Fearn, 2018; Biswas et al.,

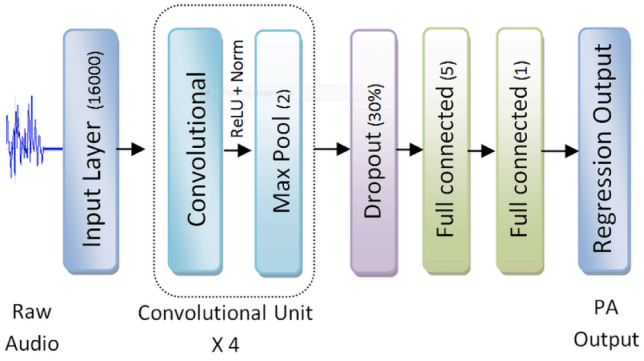


Fig. 7. CNN model for regression problems.

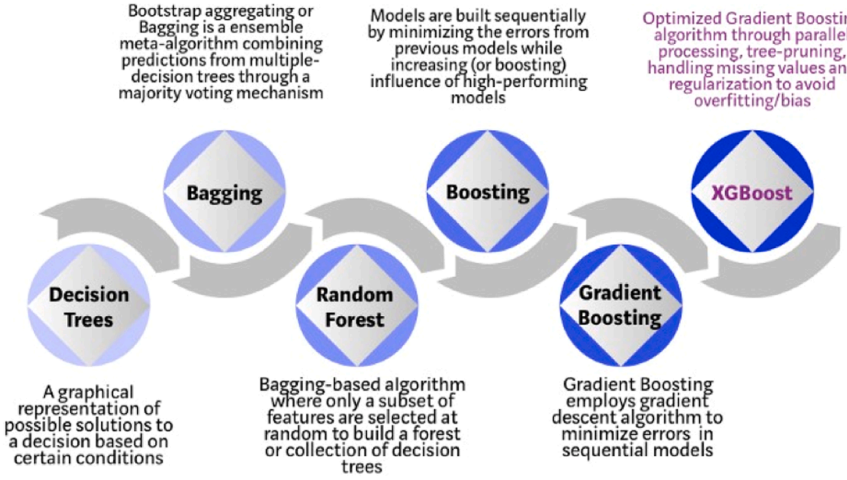


Fig. 6. XGBoost evolution from decision tree.

2024). The architecture of a CNN includes (Fig. 7):

- Convolutional Layers: These layers extract features from the input data. They perform convolutions to capture spatial hierarchies and patterns. The parameters (filters and kernel size) remain the same as in classification problems.

$$(f * x)(i,j) = \Sigma_m \Sigma_n x(i+m, j+n) f(m,n) \tag{12}$$

where x is the input, f is the filter, and * denotes the convolution operation.

- Pooling Layers: These layers reduce the spatial dimensions of the feature maps, leading to computational efficiency and invariance to small translations.

$$MaxPooling(x) = \max(x_{ij}) \tag{13}$$

- Fully Connected (Dense) Layers: After the convolutional and pooling layers, the feature maps are flattened and fed into fully connected layers, which combine the features learned by previous layers.

$$y = F(W.x + b) \tag{14}$$

where W is the weight matrix, x is the input vector, b is the bias, F is the activation function, which can be sigmoid, ReLU, etc.

- Output Layer: For regression problems, the output layer typically has a single neuron (for a single continuous value) or multiple neurons (for multi-dimensional continuous values), and it uses a linear activation function. For classification problems, CNN often uses Softmax activation, which means the number of neurons equal to the number of classes.

$$Softmax(z_i) = \frac{e^{z_i}}{\sum_{j=1}^{K} e^{z_j}} \tag{15}$$

2.7. Long-short term memory

LSTM networks are a type of Recurrent Neural Network (RNN) designed to capture long-term dependencies and handle the vanishing gradient problem. The architecture of LSTM is more complex than that of a traditional RNN, involving several components that manage the flow of information through the network (Fig. 8). LSTM architecture includes:

- Cell State (C_t): Acts as a conveyor belt that runs through the entire LSTM, maintaining information over long sequences. The cell state undergoes minor linear transformations and is crucial for preserving long-term dependencies.

- Hidden State (h_t): This is the output of the LSTM cell at each time step, which is used in the predictions and for the next cell's input.

- Gates: LSTM has three gates (input, forget, and output) that control the flow of information into and out of the cell state.

+ Forget Gate (f_t): The forget gate determines which parts of the previous cell state (C_{t-1}) should be forgotten. It uses the previous hidden state (h_{t-1}) and the current input (x_t):

$$f_t = \sigma\left(W_f . \left[h_{t-1}, x_t\right] + b_f\right) \tag{16}$$

where:

σ: sigmoid function

W_f: the matrix of weight and bias parameters

h_t: output vector of LSTM unit

x_t: input vector of LSTM unit

+ Input Gate (i_t) and Candidate Cell State $(\widetilde{C_t})$: The input gate decides what new information will be stored in the cell state.

$$i_t = \sigma(W_i . [h_{t-1}, x_t] + b_i) \tag{17}$$

$$\widetilde{C}_t = \tanh\left(W_C . \left[h_{t-1}, x_t\right] + b_C\right) \tag{18}$$

The new cell state (C_t) is calculated by combining the old cell state (C_{t-1}) and the new candidate cell state $(\widetilde{C_t})$, as controlled by the forget gate (f_t) and the input gate (i_t):

$$C_t = f_t * C_{t-1} + i_t * \widetilde{C}_t \tag{19}$$

+ Output Gate (o_t): The output gate decides what the next hidden state (h_t) will be. It first calculates the output gate activation (o_t) and then uses it to scale the updated cell state (C_t):

$$o_t = \sigma(W_o . [h_{t-1}, x_t] + b_o) \tag{20}$$

$$h_t = o_t * \tanh(C_t) \tag{21}$$

An LSTM network typically consists of several LSTM layers stacked together to form a deep network (Huy et al., 2022):

- Input Layer: Takes in the sequential data. Each input is a vector, and the input sequence has a certain number of time steps.

- LSTM Layers: One or more LSTM layers process the sequence data. These layers can be stacked to create a deep LSTM network.

- Dense (Fully Connected) Layer: Maps the output of the LSTM layers to the desired output size.

- Output Layer: For regression tasks, this usually consists of one neuron with a linear activation function.

2.8. Combined CNN-LSTM

The CNN-LSTM architecture is a combination of Convolutional Neural Networks (CNNs) and Long Short-Term Memory (LSTM) networks, designed to leverage the strengths of both models (Livieris et al., 2020). This architecture is particularly useful for tasks that involve spatial data with temporal dependencies, such as video analysis, time series forecasting from images, and activity recognition.

The CNN-LSTM architecture typically involves the following steps (Fig. 9):

- Input data preparation: The input data for CNN-LSTM models often consists of sequences of spatial data. For example, a video can be represented as a sequence of frames.

- Feature extraction with CNN: Each frame (or spatial data point) is passed through a CNN to extract spatial features. CNN applies convolutional operations to capture spatial hierarchies and patterns. The output of the CNN for each frame is a set of high-level features, often represented as a feature map or vector.

- Sequence modeling with LSTM: The sequence of feature vectors obtained from the CNN is fed into an LSTM network. The LSTM

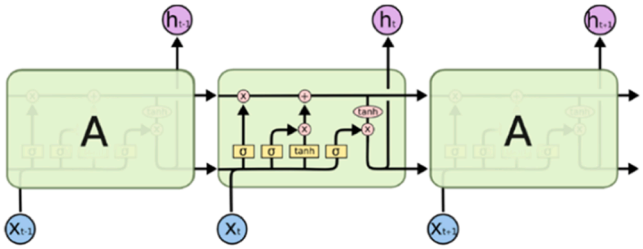


Fig. 8. LSTM architecture.

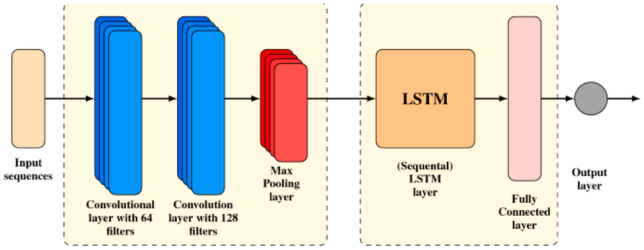


Fig. 9. Combined CNN-LSTM architecture.

processes the sequence of features to capture temporal dependencies and patterns. The LSTM maintains a memory of past information and updates it based on the new input features and its internal state.

- Prediction: The output of the LSTM network can be passed through fully connected (dense) layers to generate the final predictions. Depending on the task, this could be a classification label, a sequence of values, or another form of output.

3. Results

3.1. Fault classification

In this section, machine learning models such as SVM, DT, LR, XGBoost, and ANN are used to classify short circuit faults in the IEEE 9 Bus system (Fig. 10). This system consists of 9 buses, 3 generators, 3 loads and 3 transformers. MATLAB Simulink is used to simulate the system and generate the fault dataset (Fig. 11).

The datasets are created in both normal conditions and fault conditions. In normal conditions, the load is set at three levels: high, medium, and low. At each level, the load varies between ±5 %Pmax with the step of 1 %Pmax (Table 1). In this system, loads are located at Bus 5,7,9 (Fig. 10). Therefore, the total number dataset in normal condition is 30^3 = 27000. In fault conditions, for simplicity, load is set at only three levels with no variation (100 %, 70 %, and 50 %Pmax). There are 10 short circuit faults (AG, BG, CG, AB, BC, AC, ABG, BCG, ACG, and ABC). The fault resistance is set at 10 values (Table 2). The IEEE 9 Bus system is divided into 4 zones (Fig. 11). In zones 1,2 and 3, there are two fault positions: in the low voltage and high voltage side of the transformer. In zone 4, faults are generated in the middle of the 6 transmission lines. The total number of fault positions in all four zones is 12. Therefore, the total number of datasets in fault condition is $3^3 \times 10 \times 10 \times 12 = 32400$, and the total number of datasets in both normal and fault conditions is 27000+32400 = 59400.

To classify the fault, the datasets need to be labeled. In this part, we will label each type of faults (LG – line to ground, LL – line to line, LLG – line to line to ground, and LLL – 3 phase) in each of four zones, and one label for normal conditions. Therefore, the total number of labels is 4×4+1=17. The dataset is divided into training and testing set with the ratio of 80:20. Phase voltages and currents from Bus 1, 2 and 3 (generator) are used as input data while 17 labels are used as output classification (Table 3). Table 4 shows the example of training and testing dataset. In the fault classification problem, the output is labeled from 0 to 16, which corresponds to 17 fault classification labels. In the fault localization problem, the output is the fault position in km.

Fig. 12 represents the flow chart for the machine learning models training and evaluation, including data collection, data preprocessing, model selection, model training, model evaluation, and model deployment. Initially, data is collected through simulations using MATLAB software. During the data preprocessing phase, the collected data undergoes cleaning and transformation to ensure it is suitable for model building. This involves normalizing features, encoding categorical variables, and splitting the data into training and testing datasets. In the model selection phase, different machine learning algorithms are considered. During model training, the selected model is trained on the preprocessed data, learning patterns and relationships within the training data and adjusting its parameters to maximize prediction accuracy. After training, the model is evaluated by predicting the testing data, which was separated from the training data. Depending on the task, suitable metrics are chosen to evaluate the model's performance. If the model does not meet the predefined performance requirements, the process loops back to the model selection step for further tuning or

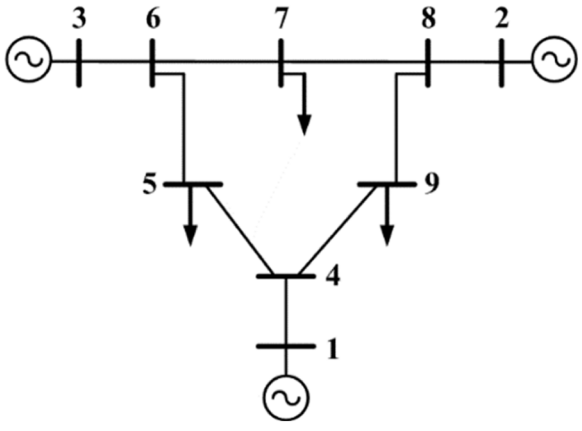


Fig. 10. IEEE 9 bus system diagram.

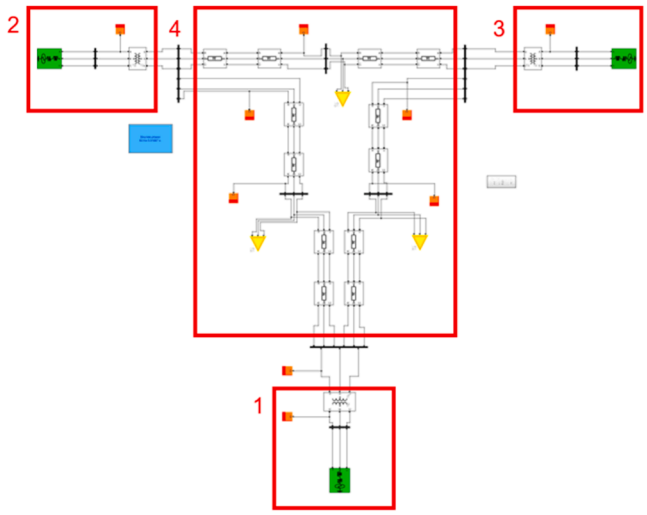


Fig. 11. IEEE 9 bus system simulation in MATLAB Simulink.

Table 1
Load variation data.

Load level	Value	Step
High	95 %–105 %	1 %
Medium	65 %–75 %	1 %
Low	45 %–55 %	1 %

Table 2
Fault resistance value.

Fault Resistance	R(Ω)	Fault Resistance	R(Ω)
Case 1	0.001	Case 6	0.5
Case 2	0.005	Case 7	1
Case 3	0.01	Case 8	5
Case 4	0.05	Case 9	10
Case 5	0.1	Case 10	20

Table 3
Fault classification labels.

Labels	Description	Labels	Description
1–1	LG in zone 1	2–1	LG in zone 2
1–2	LL in zone 1	2–2	LL in zone 2
1–3	LLG in zone 1	2–3	LLG in zone 2
1–4	LLL in zone 1	2–4	LLL in zone 2
3–1	LG in zone 3	4–1	LG in zone 4
3–2	LL in zone 3	4–2	LL in zone 4
3–3	LLG in zone 3	4–3	LLG in zone 4
3–4	LLL in zone 3	4–4	LLL in zone 4
0	Normal		

Table 4
Example of training and testing dataset.

I1A	I1B	I1C	I2A	I2B	I2C	I3A	I3B	I3C	U1A	U1B	U1C	U2A	U2B	U2C	U3A	U3B	U3C	Output
1.54	1.49	0.76	1.59	0.88	0.69	0.77	0.44	0.33	0.48	0.48	1.50	0.48	0.68	1.00	0.42	0.69	1.16	2
2.03	0.85	0.79	1.20	0.61	0.89	0.65	0.33	0.39	0.64	1.47	1.47	0.61	0.99	0.97	0.53	1.13	1.12	0
2.30	0.86	2.65	1.21	0.71	1.62	0.63	0.34	0.76	0.55	1.44	0.46	0.55	0.96	0.51	0.49	1.12	0.49	2
2.02	0.81	0.76	1.17	0.56	0.84	0.64	0.30	0.36	0.65	1.48	1.48	0.62	1.01	0.99	0.54	1.16	1.15	0
2.10	0.89	1.28	1.43	0.69	1.38	0.86	0.35	0.93	0.60	1.44	1.27	0.49	0.97	0.83	0.15	1.11	0.93	10
2.04	0.89	0.83	1.20	0.63	0.91	0.65	0.33	0.38	0.64	1.44	1.43	0.61	0.97	0.95	0.53	1.13	1.11	0
2.02	0.79	0.78	1.15	0.50	0.79	0.62	0.28	0.36	0.65	1.49	1.49	0.62	1.02	1.01	0.54	1.16	1.16	0
2.60	1.64	0.84	1.76	1.03	0.78	0.81	0.46	0.35	0.46	0.78	1.45	0.46	0.62	0.96	0.40	0.66	1.12	14
1.99	0.69	0.66	1.11	0.43	0.70	0.61	0.25	0.31	0.66	1.55	1.56	0.63	1.06	1.04	0.55	1.22	1.22	0
2.01	0.79	0.75	1.16	0.52	0.81	0.63	0.29	0.36	0.65	1.49	1.50	0.62	1.02	1.00	0.54	1.17	1.16	0
2.04	0.87	0.83	1.21	0.62	0.91	0.65	0.33	0.41	0.64	1.45	1.45	0.61	0.98	0.96	0.53	1.11	1.11	0
2.03	0.90	0.84	1.18	0.63	0.89	0.64	0.19	0.29	0.64	1.47	1.45	0.61	1.00	0.96	0.54	1.15	1.09	10
2.24	1.58	0.78	1.51	1.07	0.78	0.79	0.66	0.38	0.57	0.85	1.46	0.52	0.63	0.95	0.42	0.50	1.12	15
2.07	0.84	1.31	2.12	0.67	1.69	0.73	0.34	0.65	0.60	1.47	1.04	0.39	0.97	0.42	0.39	1.12	0.66	6
2.02	0.81	0.75	1.17	0.55	0.83	0.63	0.30	0.35	0.65	1.49	1.49	0.62	1.01	0.99	0.54	1.16	1.15	0
1.54	1.49	0.76	1.59	0.88	0.69	0.77	0.44	0.33	0.48	0.48	1.50	0.48	0.68	1.00	0.42	0.69	1.16	2
2.03	0.85	0.79	1.20	0.61	0.89	0.65	0.33	0.39	0.64	1.47	1.47	0.61	0.99	0.97	0.53	1.13	1.12	0

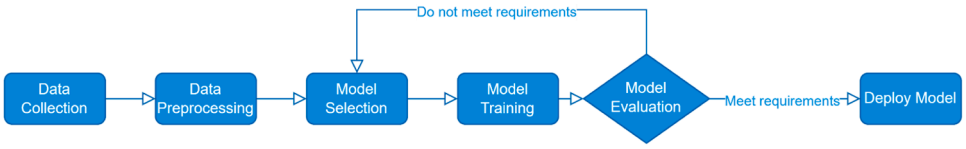


Fig. 12. Flow chart of training machine learning model.

selection of a different model. Once the model meets the performance criteria, it is deployed into a production environment where it can make predictions on new, unseen data. This step involves integrating the model with existing systems and ensuring it operates efficiently in real-time applications.

Fig. 13 illustrates how a machine learning model operates after deployment. The process starts with the collection of data required for the machine learning task. Following data collection, the raw data is normalized to ensure it is suitable for the deployed model. Once the data is collected and preprocessed, the deployed model is called. The loaded model is then applied to the preprocessed data for prediction. The results of the model's predictions can be further analyzed, visualized, or used for decision-making in the relevant application domain.

The confusion matrix is often used to evaluate the performance of a classification algorithm. It provides a detailed breakdown of the actual versus predicted classifications made by a model, allowing for a comprehensive analysis of its accuracy.

For a binary classification problem, the confusion matrix is a 2×2 matrix is illustrated in Fig. 14.

Where:

True Positives (TP): The number of instances correctly predicted as positive.

True Negatives (TN): The number of instances correctly predicted as negative.

False Positives (FP): The number of instances incorrectly predicted as positive.

False Negatives (FN): The number of instances incorrectly predicted as negative.

There are several important performance metrics of the ML algorithms which can be calculated from confusion matrix:

- Accuracy: the proportion of correct predictions (both positive and negative) out of the total predictions made.

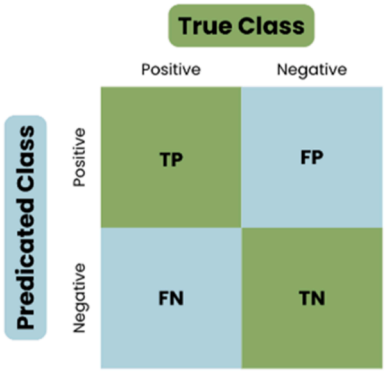


Fig. 14. : Confusion matrix of binary classification.

$$accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{22}$$

- Precision (also known as Positive Predictive Value): measures the proportion of correctly predicted positive instances out of all instances predicted as positive.

$$\text{Precision} = \frac{TP}{TP + FP} \tag{23}$$

- Recall (also known as Sensitivity or True Positive Rate): measures the proportion of correctly predicted positive instances out of all actual positive instances.

$$\text{Recall} = \frac{TP}{TP + FN} \tag{24}$$



Fig. 13. Flow chart of machine learning model implementation in real-time.

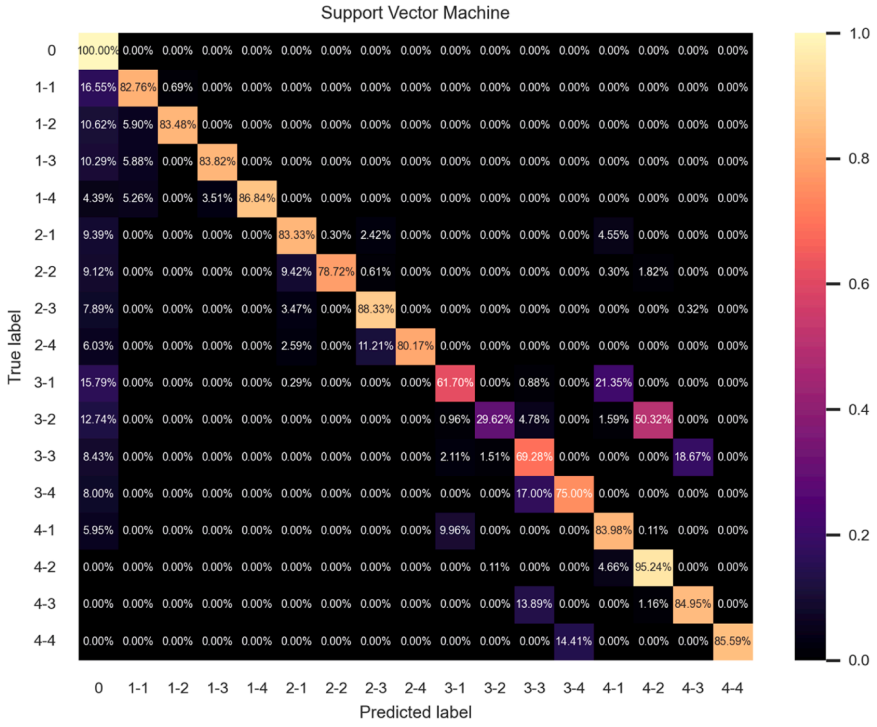
- F1-Score: is the harmonic mean of Precision and Recall, providing a single metric that balances both metrics.

$$F1-Score = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{25}$$

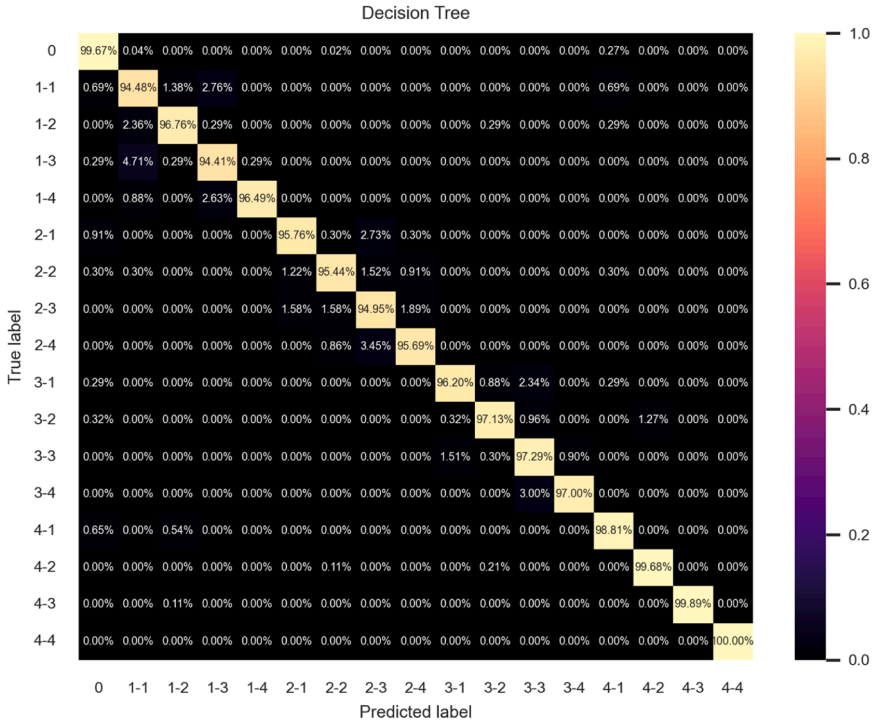
- Specificity (also known as True Negative Rate): measures the proportion of correctly predicted negative instances out of all actual negative instances.

$$Specificity = \frac{TN}{TN + FP} \tag{26}$$

For multi-class classification, the confusion matrix is an NxN matrix

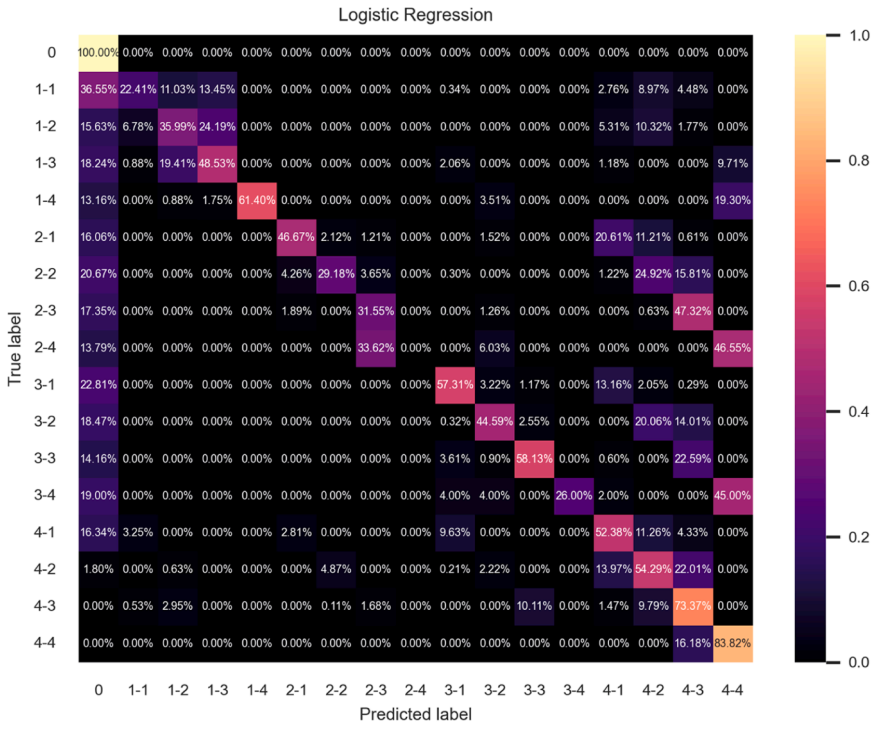


(a) Support Vector Mahine

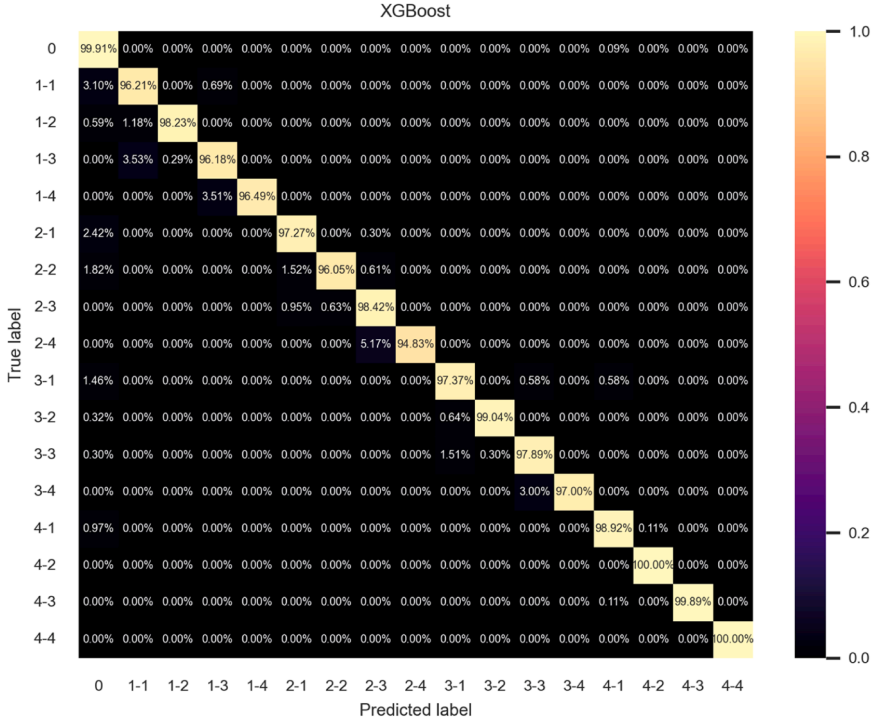


(b) Decision Tree

Fig. 15. Confusion matrix of machine learning algorithms.



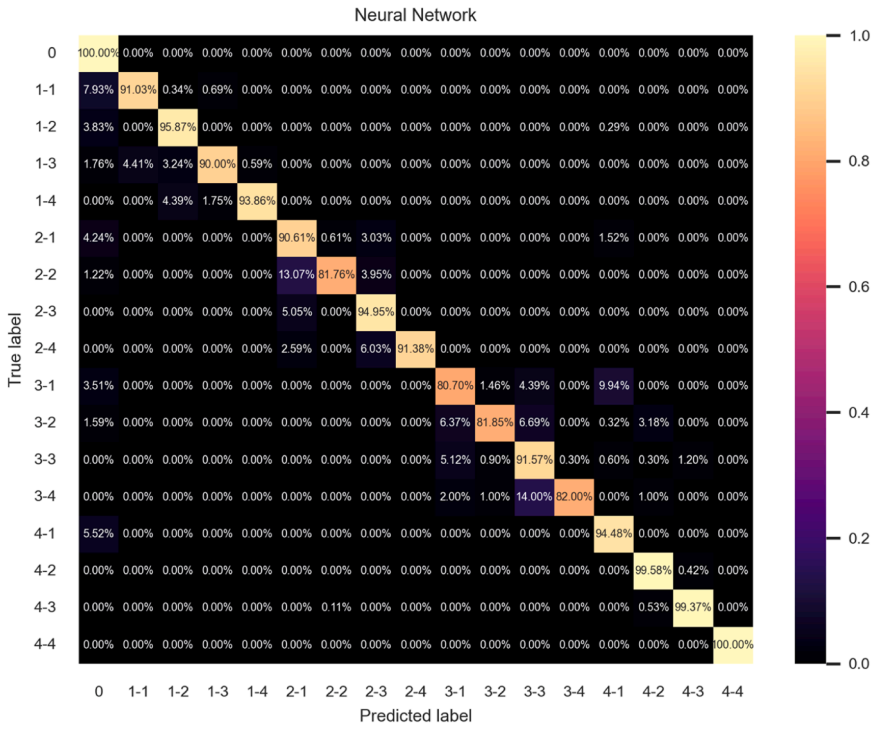
(c) Logistic Regression



(d) XGBoost

Fig. 15. (*continued*).

where N is the number of classes (labels). Each cell C(i,j) in the matrix represents the number of instances of class i^{th} that were predicted as class j^{th}. The diagonal cells represent correct predictions, while off-diagonal cells represent misclassifications. In this research, the confusion matrix dimension is 17×17 since there are 17 labels. The confusion matrix of the SVM, DT, Logistic Regression, XGBoost, and ANN are illustrated in Fig. 15. The horizontal axis is the predicted labels from the models, while the vertical axis is the actual labels. The diagonal elements of the matrix represent the percentage of correct predictions, while the off-diagonal elements represent the percentage of incorrect predictions. For example, in the SVM models, there are 9.96 % LG faults in zone 4 incorrectly classified as LG faults in zone 3 and 4.78 % LL faults



(e) Artificial Neural Network

Fig. 15. (*continued*).

in zone 3 incorrectly classified as LLG faults in zone 3.

The accuracy of the algorithms is calculated as:

$$accuracy = \frac{correct\ preditions}{all\ predictions} \tag{27}$$

The accuracy comparison between machine learning models is illustrated in Table 5. All algorithms except Logistic Regression can achieve very high accuracy (>91 %). The Logistic Regression is a simple model and often used for binary classification problems. However, it may not work well for large and complex datasets. The XGBoost algorithm has the highest accuracy of 99.82 %. XGBoost provides built-in methods to assess feature importance, which can be very useful for understanding the model and for feature selection. In addition, XGBoost also includes built-in L1 (Lasso) and L2 (Ridge) regularization, which helps prevent overfitting by penalizing large coefficients. Therefore, XGBoost often delivers high performance in terms of accuracy and other metrics compared to other algorithms.

3.2. Fault localization

In this section, CNN, LSTM, and combined CNN-LSTM are used to localize the fault position on transmission lines in IEEE 9 Bus system. Line 4–5 is selected as the line where faults occur. The length of this line is 90 km and faults are generated at each km in the line. There are ten short circuit faults (AG, BG, CG, AB, BC, AC, ABG, BCG, ACG, and ABC), three load levels with no variation (100 %, 70 %, and 50 %Pmax), 10 fault resistance values (Table 2). Therefore, the number of fault dataset in line 4–5 is: $3^3 \times 90 \times 10 \times 10 = 243000$.

Table 5
Metric comparison between classification models.

Metrics	SVM	DT	LR	XGBoost	ANN
Accuracy	89.69 %	98.59 %	73.77 %	99.11 %	96.40 %
Training time	18.17 s	4.12 s	2.62 s	8.82 s	115.53 s

Like fault classification, the phase current and voltage at Bus 1,2 and 3 (generator) are used as the input data. The distance from the fault location to Bus 4 is used as the output data. 70 % of the dataset is used for training, and the remaining 30 % is used for testing.

In regression problems, the goal is to predict a continuous output variable, and in this case, the fault position (km). To evaluate the performance of a regression model, MAE, MSE and MAPE metrics are often used to measure the difference between the predicted values and the actual values.

- Mean Absolute Error (MAE): calculates the average absolute difference between the actual values (y_i) and the predicted values ($\hat{y}_i$). It gives a linear score which means all individual differences are weighted equally in the average. Lower MAE values indicate better model performance.

$$MAE = \frac{1}{n}\sum_{i=1}^{n}|\hat{y}_i - y_i| \tag{28}$$

- Mean Squared Error (MSE): calculates the average squared difference between the actual and predicted values. Squaring the differences gives more weight to larger errors. MSE is often used as loss function in regression models. Lower MSE values indicate better model performance.

$$MSE = \frac{1}{n}\sum_{i=1}^{n}(\hat{y}_i - y_i)^2 \tag{29}$$

- Mean Absolute Percentage Error (MAPE): calculates the average absolute percentage difference between the actual and predicted values. Lower MAPE values indicate better model performance.

$$MAPE = \frac{1}{n}\sum_{i=1}^{n}\left|\frac{\hat{y}_i - y_i}{y_i}\right| \times 100 \tag{30}$$

Fig. 16 illustrates the MSE and MAE of each algorithm in both training (blue curve) and testing (orange curve) dataset. The horizontal axis represents the number of epochs (or iterations) during the training

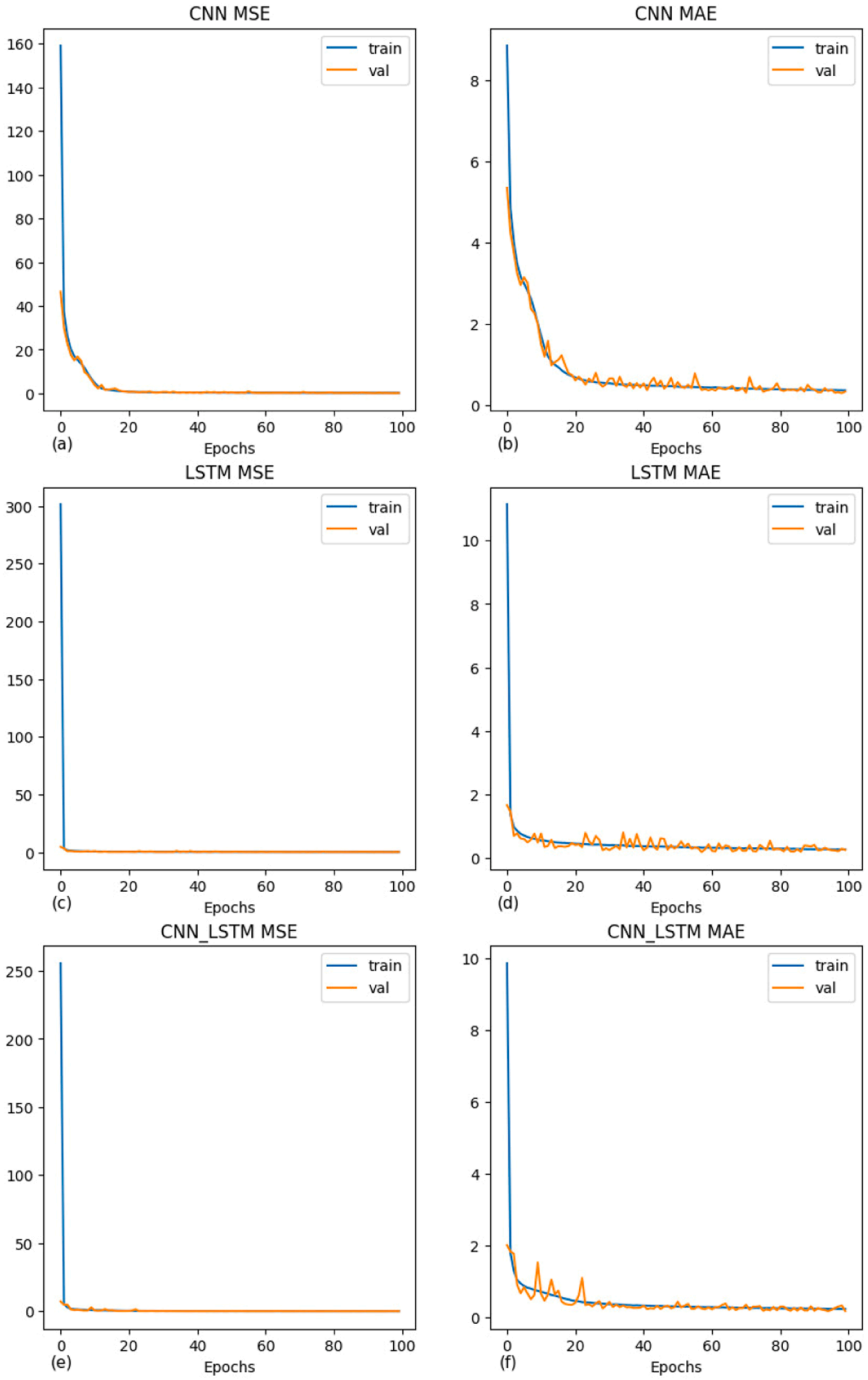


Fig. 16. Comparison between MSE and MAE of CNN, LSTM and CNN-LSTM algorithms.

process, indicating the progress of the model training.

In this research, the MSE is used as the loss function of all regression models, therefore, the MSE curve doesn't fluctuate during training process. The MAE curve, on the other hand, observes fluctuation during the first stage of the training process. This comes from several factors. The first one is random initialization. At the start of training, model parameters (such as weights in neural networks) are typically initialized randomly. This can lead to initially unpredictable predictions, causing fluctuations in MAE. The second factor is batch variability. During training, especially with stochastic gradient descent (SGD) or mini-batch gradient descent, the model is updated based on a subset of the training data. The composition of these batches can vary, causing the MAE to fluctuate. In addition, the complexity of the data is an important factor affecting the training process. If the data has high variability or noise, the model might initially struggle to find patterns, leading to fluctuations in the MAE as it starts to learn the underlying structure of the data. As the number of epochs increases, the model parameters progressively adjust to better fit the training data. Each epoch typically leads to a smaller and more refined step towards the minimum of the loss function, which reduces the error. As observed, optimized models can be obtained after 100 epochs.

Metric comparison of CNN, LSTM and CNN-LSTM is shown in Table 6. All algorithms show the great accuracy (MAPE $<$ 1,88 %, MAE $<$ 0,29 km). The LSTM provides better results than the CNN algorithm. The combined CNN-LSTM provides the best results with MAPE $<$ 1 % and MAE $<$ 0,16 km. However, it takes the longest training time.

Table 6
Metric comparison between faults localization models.

Metrics	CNN	LSTM	CNN-LSTM
MSE	0.164	0.072	0.053
MAE	0.293	0.183	0.158
MAPE %	1.884	1.315	0.996
Training time	42 minutes	116 minutes	182 minutes

4. Conclusions

This paper presents a novel approach to classify and localize faults in transmission lines based on machine learning algorithms. The IEEE 9 Bus system is simulated in MATLAB Simulink to generate large dataset for model training. Fault datasets are generated with various parameters such as fault types, fault resistance, load level, fault position, to mimic the practical condition. 59400 datasets are created for fault classification, while 243000 datasets are created for fault localization. The results show that machine learning models can classify and localize faults with very high accuracy. In classification task, XGBoost can classify faults with the accuracy of 99.82 %. In regression task, the combined CNN-LSTM can localize faults with MAPE $<$ 1 % and MAE $<$ 160 m in a 90 km transmission line. Classifying and localizing faults in transmission lines correctly plays a crucial role to enhance the power system stability and reliability. In future works, the error in fault data will be considered to mimic the current and voltage transformers error in practical. In addition, the models will be enhanced to reduce the training time, especially the regression models.

CRediT authorship contribution statement

Vu Hoai Giang: Writing – review & editing, Conceptualization. **Nguyen Trong Khiem:** Validation, Methodology, Data curation, Conceptualization. **Minh Quoc Nguyen:** Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Conceptualization.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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